

Research Interests

My research interests lie in group (co)homology of congruence subgroups of arithmetic groups and virtual duality groups. I am also interested in questions related to homological stability for families of groups.

Research Papers

Top-dimensional rational cohomology of $\Gamma_{0,n}^+(p)$: arXiv:2605.23526

Education

2021-Present **Ph.D in Mathematics**, *University of Oklahoma*

Advisor: Peter Patzt

2019-2021 **M.S. in Mathematics**, *American University of Beirut*

Advisor: Hazar Abu-Khuzam

Thesis Title: Structure of certain rings with conditions on nonperiodic elements and certain Von-Neumann II-regular rings with prime centers

2015-2018 **B.S. in Mathematics**, *Lebanese University*

Honors and Awards

2026 **MGSA Good Mentor Award**, *University of Oklahoma*

2026 **Thomas J. Hill Graduate Fellowship in Mathematics**, *University of Oklahoma*

2026 **T. Jay Thompson scholarship ward**, *University of Oklahoma*

2026 **Armstrong Memorial scholarship award**, *University of Oklahoma*

2026 **SHS-OG&E Energy Corp scholarship award**, *University of Oklahoma*

2026 **Dodge Family Completion Fellowship award**, *University of Oklahoma*

2025 **David and Judi Proctor Scholarship**, *University of Oklahoma*

2025 **Robert Lee Loshbough Endowed Scholarship Fund**, *University of Oklahoma*

2025 **Arts and Sciences - 31171**, *University of Oklahoma*

2024 **Summer Graduate Fellowship**, *University of Oklahoma*

2023 **Conrad and Ellen Masterson Fellowship in Mathematics**, *University of Oklahoma*

2022 **C. Eugene Springer Mathematics Scholarship**, *University of Oklahoma*

2021-2025 **Fellowship from the Mathematics department**, *University of Oklahoma*

2021-Present **Graduate Teaching Assistantship**, *University of Oklahoma*

2019-2021 **Graduate Assistantship**, *American University of Beirut*

Talks and Presentations

May 2026 **Cohomology of congruence subgroup $\Gamma_{0,n}(p)$** , *International Seminar on Quantization, Geometry, and Dynamics, University of Texas Rio Grande Valley (virtual)*

- Apr. 2026 **Top-dimensional cohomology of congruence subgroup $\Gamma_{0,n}(p)$** , *Topology seminar, Purdue University*
- Feb. 2026 **Top-dimensional cohomology of congruence subgroup $\Gamma_{0,n}(p)$** , *Topology seminar, University of Michigan*
- Nov. 2025 **Borel–Serre duality (expository)**, *Student Topology Seminar, University of Oklahoma*
- Oct. 2025 **Top-dimensional rational cohomology of the congruence subgroup $\Gamma_{0,n}(p)$** , *Geometry and Topology Seminar, University of Oklahoma*
- Oct. 2025 **Top-dimensional rational cohomology of the congruence subgroup $\Gamma_{0,n}(p)$ (poster presentation)**, *Perspectives on the cohomology of general linear groups, University of Münster*
- Jul. 2025 **Double Steinberg Coinvariants, Scissors Congruence and K-theory Conference**, *University of Pennsylvania*
- Nov. 2023 **High Dimensional Cohomology of Arithmetic Groups (expository)**, *The 2023 Binghamton University Graduate Combinatorics, Algebra, and Topology Conference, Binghamton University*
- Dec. 2022 **Introduction to Spectral Sequences (expository)**, *Student Presentation Seminar, University of Oklahoma*

Teaching experience

Instructor of record:

- SP26 **Differential & Integral Calculus II (HONORS)**, *University of Oklahoma*
- FA25 **Functions and Modeling for Business, Life and Social Sciences. (large section ~ 80 students)**, *University of Oklahoma*
- SP24 **Calculus I for Business, Life and Social Sciences**, *University of Oklahoma*
- FA23 **Functions and Modeling for Business, Life and Social Sciences.**, *University of Oklahoma*
- SU23 **Calculus & Analytics Geometry IV**, *University of Oklahoma*
- SP23 **Functions and Modeling for Business, Life and Social Sciences.**, *University of Oklahoma*
- SU22 **Calculus & Analytics Geometry I**, *University of Oklahoma*

Discussion Leader:

- SP25 **Differential & Integral Calculus II (HONORS)**, *University of Oklahoma*
- FA24 **Differential & Integral Calculus II (HONORS)**, *University of Oklahoma*
- FA22 **Differential & Integral Calculus I**, *University of Oklahoma*
- SP22 **Calculus and Analytics Geometry I**, *University of Oklahoma*

Grader:

- SP25 **Topology II**, *University of Oklahoma*
- FA24 **Topology I**, *University of Oklahoma*
- FA22 **Introduction to Analysis I**, *University of Oklahoma*
- FA21 **Calculus & Analytics Geometry IV**, *University of Oklahoma*
- FA21 **Differential & Integral Calculus III (HONORS)**, *University of Oklahoma*
- FA20 **Rings & Modules I**, *American University of Beirut*

FA19,SP20 **Introduction to Analysis I**, *American University of Beirut*

Conferences and Workshops attended

- Jun. 2026 **Topology Students Workshop**, *Vanderbilt University*
May. 2026 **Configuration spaces summer school**, *University of Michigan*
Jun. 2025 **Polylogarithms, homology of linear groups, and Steinberg modules**, *SwissMAP Research*, Station in Les Diablerets
May. 2025 **Summer school: Homological aspects of groups and their classifying spaces**, *SwissMAP Research*, Station in Les Diablerets
Feb. 2025 **Redbud Topology Conference Spring 2025**, *University of Arkansas*
Jul. 2024 **Collaborative Workshop in K-theory and Scissors Congruence**, *Vanderbilt University*
Oct. 2023 **The South Central Topology Conference II**, *New Mexico State University*
Oct. 2023 **Redbud Topology Conference Fall 2023**, *University of Arkansas*
Jul. 2023 **Stability in Topology, Arithmetic, and Representation Theory 2023**, *Purdue University*
Jun. 2023 **Master class : Homology of groups and functors**, *Université de Lille*
April 2023 **Redbud Topology Conference Spring 2023**, *University of Oklahoma*
Feb. 2023 **The South Central Topology Conference II**, *Texas State University*
Nov. 2022 **The 2022 Binghamton University Graduate Combinatorics, Algebra, and Topology Conference**, *Binghamton University*
Oct. 2022 **Redbud Topology Conference Fall 2022**, *University of Arkansas*
April 2022 **Texas-Oklahoma Representations and Automorphic forms XI**, *Oklahoma State University*

Service and Outreach

- 2025-2026 **Treasurer of the OU AWM Chapter**, *University of Oklahoma*
Spring 2025 **Mentor in the Directed Reading Program**, *On the proof of Abel and Ruffini*, *University of Oklahoma*
2024-2025 **Graduate Student Senate**, *University of Oklahoma*
2024-2025 **Outreach of the OU AWM Chapter**, *University of Oklahoma*
April 2024 **Co-organizer of the OU Graduate Math Fest**, *University of Oklahoma*
2023-2024 **Co-founder & President of the OU AWM Chapter**, *University of Oklahoma*
2023-2024 **Secretary of the Math Graduate Student Association**, *University of Oklahoma*
2022-2023 **Member of the Graduate Advocacy Committee**, *University of Oklahoma*